



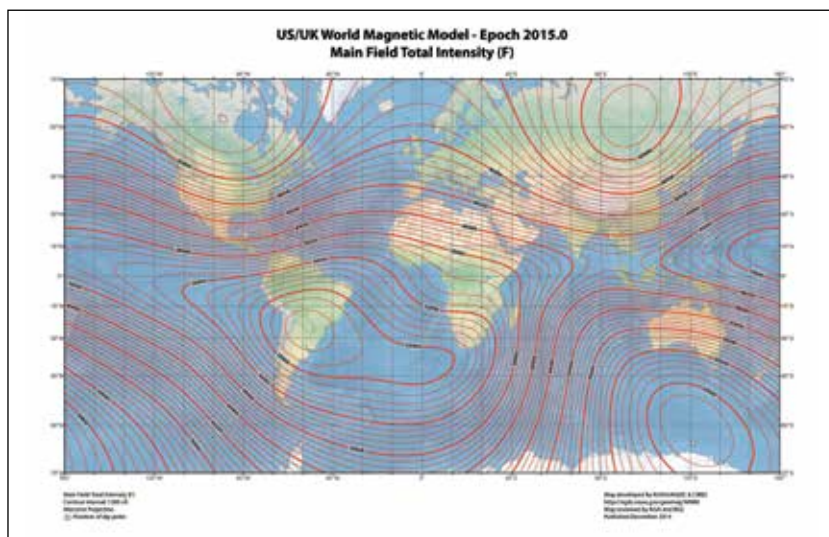
Galaxy S9 and S9 plus launch

Samsung is reportedly readying for a huge launch. The smartphones are apparently making their way to MWC 2018. <http://dgit.in/SamGS9>



OnePlus 5T to get Oreo Update

OnePlus to roll out Android Oreo update this month for 5 and in December for 5T <http://dgit.in/OneP5up>



Just an example of the magnetic field lines in 2015

This is a pair of electrons in a molecule with opposite spin – quantum entangled electrons.

So when a photon is absorbed by one of the electrons, it jumps to a higher state (or a higher orbit), and this makes the electron pair more susceptible to the magnetic field of the earth. The longer the electron stays in the higher orbit, the more accurate the reading. This reading is done when the electron finally jumps back to its original state and lets off a small quanta of energy, which is dependent on how perturbed the electron was by the magnetic field. Birds may be sensing the magnetic lines of the earth and actually seeing them the way we draw them on maps ourselves!

Honestly, this could still be proven wrong, but all experiments done so far have upheld the theory. So, birds? That's what quantum biology is about? Actually no. There's even more evidence emerging that is pointing to quantum effects being responsible for all life as we know it!

PHOTOSYNTHESIS

Although most life isn't photosynthetic, and some doesn't even depend on the process at all, not even for food, it's pretty obvious that life on earth thrives because of its ability to use sunlight to make food. Plants and

bacteria make food from sunlight, then they are food for herbivores and other bacteria, which in turn are food for carnivores, and everything eventually dies and becomes plant food, or bacteria food, and the circle of life is complete. Take away photosynthesis, and earth would be a very barren place indeed, with (maybe) a few extremophiles living near undersea volcanoes, and that's about it.

For decades we've felt that we understand all there is to know about photosynthesis. All of us have learnt that sunlight falls on leaves that contain chlorophyll and then something happens and the plant is able to make food by using sunlight and carbon dioxide to "eat" carbon and give out oxygen... and it's why we're all here... so plant more trees... Sound familiar?

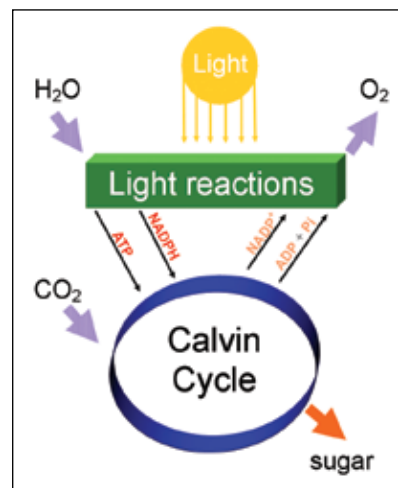
If you're a little more into biology you will know the explanation that involves chloroplasts and photons and reaction centres. The shortest version we can muster for the purpose of this article is that photosynthesis happens in light-dependent and light-independent reactions, and it's the light independent one that actually breaks apart carbon dioxide. The light dependent reaction breaks water into hydrogen and releases oxygen. Thus, the whole "plants use sunlight to extract oxygen from carbon dioxide" idea can be

considered wrong depending on how much of a stickler you are for accuracy.

For the light dependent reaction, photons hit chlorophyll molecules, which excites an electron in the molecule. These excited electrons are transported away, and the chlorophyll molecule replaces its missing electron with one from a water molecule, which ends up breaking up the molecule into hydrogen ions (protons) and oxygen. The hydrogen ions then attach to some complex compounds with the help of enzymes, and all of these undergo a light independent chemical reaction that we know as the Calvin cycle – named after American chemist Melvin C Calvin who mapped the cycle using radioactive carbon. Although far more complex a reaction than we can explain here, what this cycle essentially does is absorb and use carbon dioxide from the atmosphere.

The reason for that crash course in photosynthesis is to explain that the excited electron (called an exciton) jumps from one chlorophyll molecule to another, making a pathway to what's called the reaction center, where all the chemical reactions occur. Think of it as how you eat with your mouth, but digest with your stomach, because it would be inefficient to have stomachs where our mouths are.

However, transport of electrons uses up energy, and there's no one path that this electron can take because of



Oversimplified reaction of photosynthesis